

Does SAM Help a Small MLP Under Label Noise? A Controlled CPU Study Against SGD and Weight Decay

Quartz-46Y
STEM seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/sam-vs-sgd-label-noise-mlp>

Abstract—We compare SGD, SGD with weight decay (SGD+WD) and sharpness-aware minimisation (SAM) on a three-layer MLP trained on scikit-learn digits with symmetric label noise (0, 20, 40%) and on a noisy two-spirals task, using five seeds, hyper-parameters selected on a noisy validation split with separate tuning seeds, and a fixed budget of 150 epochs (SAM therefore uses twice the gradient evaluations). SAM clearly beats plain SGD under noise (clean test accuracy 0.938 vs 0.799 at 20% noise and 0.901 vs 0.625 at 40%), but most of this gain is also obtained by a tuned weight decay: the margin over SGD+WD is 0.008 at 20% noise (paired $p = 0.049$, Welch $p = 0.29$, so we call it inconclusive) and 0.027 at 40% (both $p < 0.05$ uncorrected; one of five seeds is negative and the paired test does not survive multiple-comparison correction). On two-spirals SAM does not help. On noisy digits accuracy rises with the radius ρ up to $\rho = 0.5$ and collapses to chance at $\rho = 1.0$ (no value in between was tested); a random-direction perturbation of the same radius behaves like SGD. A fixed-norm loss-increase sharpness is only weakly or inconsistently correlated with the generalisation gap (pooled Spearman 0.07–0.24 over the two sharpness measures and two gaps, negative on spirals); on noisy digits adversarial sharpness ranks SGD, which memorises the flipped labels, as flatter than the better-generalising SAM and SGD+WD, whereas random-direction sharpness ranks SAM flattest. This is a small, single-architecture CPU study with reimplemented optimisers; we make no claim beyond this scale.

Index Terms—sharpness-aware minimisation, label noise, weight decay, sharpness, generalisation gap, small MLP

I. Introduction

Sharpness-aware minimisation (SAM) [1] trains on the worst-case loss within a small ball around the weights, and has been reported to improve generalisation and, in particular, to be robust to noisy labels. This is usually motivated by the idea that flat minima generalise better [2], yet the evidence that sharpness explains generalisation is mixed [3], [4], and recent analyses suggest that SAM’s behaviour under label noise is not simply a flatness effect [5], [6]. Networks that can fit random labels [7] make label noise a natural stress test: an optimiser must avoid memorising the flipped labels.

Quartz-46Y is the codename of an autonomous study, not a person. Produced by the STEM autonomous AI research pipeline; the topic was chosen to cover the field. Every reported number comes from logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

We ask three questions at a scale that fits a laptop CPU, with a validation-selected ρ (6 values) for SAM, a validation-selected λ (4 values) for SGD+WD, and a shared, untuned learning rate: H1: does SAM improve clean test accuracy over SGD and over SGD with tuned weight decay under label noise? H2: how sensitive is SAM to the neighbourhood size ρ , and is there an interior optimum? H3: does a measured sharpness (loss increase under a fixed-norm weight perturbation) track the generalisation gap?

Our contributions are modest and empirical: (i) a three-optimiser comparison on digits with 0/20/40% symmetric label noise and on a two-spirals task, with 5 evaluation seeds and hyper-parameter selection on separate tuning seeds; (ii) a ρ sweep, a weight-decay sweep and two ablations (random-direction perturbation, SAM+WD); (iii) a sharpness-gap correlation analysis over 260 runs. Our headline finding is that SAM beats plain SGD by a wide margin under noise but only narrowly, and not robustly, beats SGD with weight decay, and that fixed-radius sharpness does not track the gap across optimisers.

II. Related Work

SAM and variants. SAM [1] minimises $\max_{\|e\| \leq \rho} L(w+e)$ with a one-step approximation of the inner maximum. ASAM [8] makes the neighbourhood adaptive to parameter scale, addressing the dependence of fixed-radius sharpness on re-parameterisation, and ESAM [9] reduces SAM’s cost of two gradient evaluations per step. We reimplement only the original SAM; the scale dependence that ASAM addresses matters for our sharpness measure (Sec. VII).

Understanding SAM. Andriushchenko and Flammarion [6] analyse the implicit bias of SAM (diagonal linear networks, plus convergence results) and argue that PAC-Bayes and flat-minima explanations are incomplete. Wen et al. [10] pin down which notion of sharpness SAM actually regularises. For label noise specifically, Baek et al. [5] report that SAM’s largest gains occur under label noise, that the peak occurs with early stopping rather than at convergence, and decompose the robustness into a logit effect (up-weighting clean examples) and a Jacobian effect.

We use no early stopping, so our results concern the end of training only.

Sharpness and generalisation. Keskar et al. [2] associated large-batch training with sharp minima and a larger generalisation gap; Jiang et al. [3] compared over 40 complexity measures across thousands of networks; Andriushchenko et al. [4] report that (adaptive) sharpness does not correlate well with generalisation across modern settings, but rather with training parameters such as the learning rate. Zhang et al. [7] showed that standard networks fit random labels, which motivates our measurement of how many flipped labels each optimiser memorises.

III. Method

Model and data. A ReLU MLP f_w with two hidden layers of width 128 (digits: 64–128–128–10; spirals: 2–128–128–2) is trained with cross-entropy $L(w) = \frac{1}{n} \sum_i \ell(f_w(x_i), \tilde{y}_i)$ on labels $\tilde{y}_i$ that were flipped with probability η to a uniformly chosen other class.

SGD (reimplemented, reference). With momentum $\mu = 0.9$ and learning rate η_t (cosine decay), $v_{t+1} = \mu v_t + g_t$, $w_{t+1} = w_t - \eta_t v_{t+1}$, where $g_t = \nabla_w L_{B_t}(w_t)$ is the minibatch gradient.

SGD+WD. The same update with $g_t = \nabla_w L_{B_t}(w_t) + \lambda w_t$ (L2 penalty $\frac{\lambda}{2} \|w\|^2$).

SAM (reimplemented). Following [1], the inner maximum is approximated at each step by one normalised-gradient ascent step,

$$\hat{\epsilon}_t = \rho \frac{\nabla_w L_{B_t}(w_t)}{\|\nabla_w L_{B_t}(w_t)\|_2}, \quad g_t = \nabla_w L_{B_t}(w + \hat{\epsilon}_t)|_{w=w_t}, \quad (1)$$

and g_t replaces the gradient in the momentum update above (no weight decay, $\lambda = 0$). The norm is global over all parameters. A step costs two forward/backward passes.

Ablations. SAM random direction: $\hat{\epsilon}_t = \rho u / \|u\|_2$ with $u \sim \mathcal{N}(0, I)$, same radius. SAM+WD: SAM with $\lambda > 0$.

Sharpness and gap measures. At the end of training we measure, on the full noisy training set with a fixed perturbation norm $r = 0.5$ for all systems,

$$S_{\text{adv}} = L\left(w + r \frac{\nabla L(w)}{\|\nabla L(w)\|}\right) - L(w), \quad (2)$$

$$S_{\text{rand}} = \frac{1}{20} \sum_{k=1}^{20} L\left(w + r \frac{u_k}{\|u_k\|}\right) - L(w). \quad (3)$$

The generalisation gap is $\text{gap_acc} = \text{acc}_{\text{train, noisy labels}} - \text{acc}_{\text{test, clean}}$ (also $\text{gap_loss} = L_{\text{test}} - L_{\text{train}}$). Note that against noisy labels a model that does not memorise has a train accuracy below the clean test accuracy, so the gap becomes negative. memorised is the fraction of flipped training samples predicted with their (wrong) noisy label.

IV. Experimental Setup

Tasks. digits: scikit-learn digits (1797 images, 64 features scaled to $[0, 1]$), a seed-dependent random split into 600 train, 300 validation and 897 test images. Symmetric label noise $\eta \in \{0, 0.2, 0.4\}$ is applied to train and

validation labels; test labels are clean. spirals: a two-class, 1.5-turn spiral with Gaussian input noise, 400/200/400 train/val/test points from the same generator, $\eta = 0.2$. Task names are digits_n η and spirals_n0.2.

Training. All systems (including the ablations): momentum 0.9, base learning rate 0.1 with cosine decay to 0, batch 100, 150 epochs (900 steps for digits), final-epoch model, no early stopping, no augmentation. The budget is matched in epochs, not in gradient evaluations: SAM and its ablations use two per step. The learning rate was set once from a smoke test on SGD and not tuned per system (Sec. VII). The code is the same for all systems (method/run.py); the sharpness is measured with identical settings for all.

Hyper-parameter selection. For each task, SAM’s $\rho \in \{0.02, 0.05, 0.1, 0.2, 0.5, 1.0\}$ (6 values) and SGD+WD’s $\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}, 5 \cdot 10^{-2}\}$ (4 values) were evaluated on three tuning seeds (100–102, 120 runs) and the value with the highest mean noisy validation accuracy was chosen (Table I). The tuning seeds differ from the evaluation seeds, but on digits every seed re-splits the same pool of 1797 images, so the tuning train/validation images overlap the test images of the evaluation seeds: the selection data are not disjoint from the evaluation test data (the spirals data are freshly generated per seed). This applies equally to SAM and SGD+WD. Plain SGD has nothing to tune. The ablation SAM+WD simply reuses both selected values and was not tuned jointly.

Evaluation. Seeds 0–4 (5 seeds; the seed fixes the split, noise, initialisation and batch order, so seeds are paired across systems). The main group has 60 runs; ρ and λ sweeps reuse the evaluation seeds with the non-selected values (100 and 60 runs; the selected value comes from the main group), and the ablation group has 40 runs. Statistical tests are Welch and paired t -tests on the 5 per-seed test accuracies (rh compare); with $n = 5$ they have little power. The registered tests are uncorrected; Table III also lists Holm-adjusted paired p -values over its 8 comparisons. The Spearman correlations, the Holm-adjusted p -values and the per-seed differences are computed from the logged runs by experiments/derive.py and logged as 22 analysis rows (group derived) of the run registry.

Compute. Apple M1 Pro CPU, 2 threads, PyTorch, on a shared machine. In the main group a digits run at 20% noise takes on average 3.5s for SGD and 5.0s for SAM, which makes two gradient evaluations per step; these are wall-clock times and, unlike the result metrics, they do not reproduce to the digit when a run is repeated. Code, logs and the run registry are in the repository.

V. Results

Table II gives the main results (mean $\pm$ std over 5 seeds) and Table III the paired comparisons of SAM against each baseline on test accuracy.

H1 (SAM beats SGD and SGD+WD under noise): supported against SGD, mixed against SGD+WD. With-

TABLE I
Hyper-parameters selected on validation (3 tuning seeds).

	SAM rho	SGD+WD wd
digits_n0.0	0.1	0.001
digits_n0.2	0.5	0.01
digits_n0.4	0.5	0.01
spirals_n0.2	0.05	0.001

out noise (digits_n0.0) SGD, SGD+WD and SAM reach 0.959, 0.962 and 0.970; the SAM margin over SGD (0.010) has paired $p = 0.00731$ but Welch $p = 0.056$, i.e. it is small and test-dependent. At 20% and 40% noise plain SGD memorises essentially all flipped labels (memorised 0.997 and 1.000 in Table II) and drops to 0.799 and 0.625; SAM reaches 0.938 and 0.901 and memorises far fewer flipped labels (0.156 at 20%), with differences of 0.139 and 0.276 (paired $p = 8.7 \times 10^{-5}$ and 3.9×10^{-5} ; the Welch p -values are in Table III). Against tuned SGD+WD (0.931 and 0.874), however, the margins are 0.0078 at 20% noise (paired $p = 0.049$, Welch $p = 0.294$; below one seed standard deviation of either system) and 0.027 at 40% (paired $p = 0.025$, Welch $p = 0.0073$). The per-seed differences SAM minus SGD+WD are 0.025, 0.038, 0.036, 0.039, -0.002 at 40% noise (one of five seeds negative) and 0.006, 0.012, 0.011, -0.002 , 0.012 at 20%. We therefore call the 20% comparison inconclusive and the 40% one supported only without multiple-comparison correction (the Holm-adjusted paired p over the 8 paired tests of Table III is 0.101), with the caveats that $n = 5$, and SAM’s ρ was selected from 6 values versus 4 for λ (both selected on a noisy validation split, so neither used the test set). Weight decay alone removes most of the damage of label noise at this scale: SGD+WD memorises 0.053 of flipped labels at 20% noise.

Where SAM fails. On two-spirals (20% noise) the model does not memorise noise at all (memorised 0.085–0.102 for every system) and the three systems are indistinguishable: SAM 0.912 vs SGD 0.919 and SGD+WD 0.914 (paired $p = 0.320$ and 0.606). With ρ larger than the selected 0.05, SAM is worse there (Table IV: 0.888 at $\rho = 0.1$, 0.767 at $\rho = 0.2$ and 0.518, i.e. chance, at $\rho = 0.5$). On this task SAM gives no measurable gain, so H1 does not hold there.

H2 (interior optimum in ρ , larger and wider useful range under noise): partly supported. Table IV and Fig. 1 show the sweep. On noise-free digits accuracy rises from 0.960 ($\rho = 0.02$) to 0.974 at $\rho = 0.2$ and then falls to 0.970 at 0.5 and chance-level (0.136) at 1.0, an interior optimum, but the gain over SGD (0.959 ± 0.008) is under two SGD seed standard deviations, and accuracy stays between 0.960 and 0.974 for ρ from 0.02 to 0.5. On noisy digits accuracy rises monotonically up to $\rho = 0.5$ (from 0.807 at $\rho = 0.02$ to 0.938 at 20% noise, from 0.629 to 0.901 at 40%) and then collapses to chance at $\rho = 1.0$ (0.103 and 0.118): the best grid value thus shifts from 0.2 without noise (0.1 and 0.5

reach 0.970 and 0.970) to 0.5 with noise, as hypothesised, but the useful range is not wider: under noise only $\rho = 0.5$ recovers most of the gain (0.2 gives 0.862 and 0.698). The optimum is only bracketed in (0.2, 1.0) by this grid; the selected value sits one grid step below the collapse and we tested no value other than 0.5 inside that interval, so we cannot say how close to the cliff it is. On spirals accuracy decreases monotonically from $\rho = 0.02$ (0.921) to chance at $\rho \geq 0.5$, so no interior optimum is observed (the registered refutation condition, a monotone curve, is met on this task). Nothing is claimed beyond these 6 grid points. The weight-decay sweep (Table V) has the same shape: $\lambda = 0.01$ is best on noisy digits (0.931, 0.874) and $\lambda = 0.05$ collapses (0.525, 0.099).

H3 (sharpness tracks the gap): not supported. The registered criterion required positive correlations pooled and within every task; the within-task ones are not all positive. Table VI gives Spearman correlations between the two sharpness measures and the two gaps. Pooled over all 260 runs the correlations are weak: 0.24 for random-direction sharpness and 0.12 for adversarial sharpness with gap_acc (0.07 for adversarial sharpness with the loss gap). Per task they differ in sign: on spirals they are negative (-0.72 , -0.34), on digits_n0.0 adversarial sharpness is negatively correlated with the loss gap (-0.65), and on noisy digits they are positive but modest (0.35–0.38, apart from adversarial sharpness at 20% noise: 0.04). Only within the SAM ρ sweep on digits do they reach 0.61–0.76 (90 runs), but there ρ moves both quantities and the collapsed runs at large ρ are included; removing collapsed runs (rows “no collapsed”: test accuracy below 0.15 on digits or 0.55 on spirals, an exploratory step defined after seeing the data) leaves per-task values between -0.69 and 0.23 (digits_n0.4: 0.10 and 0.09 for the accuracy gap). Across optimisers the adversarial measure contradicts the flatness story (the random-direction measure in Table II ranks SAM flattest, 6.15×10^{-4} and 5.39×10^{-4} vs 1.13×10^{-3} and 1.35×10^{-3} for SGD at 20% and 40% noise, and SGD+WD sharpest, 1.80×10^{-3} and 2.42×10^{-3} , so the two measures disagree about SGD): in Table II, SGD has the smallest adversarial sharpness at both noisy digits tasks (0.0320 and 0.0331 vs 0.0585 and 0.0753 for SAM) yet the largest gap (0.200 vs -0.125 at 20%). A fixed-radius loss increase also depends on the loss level and on the weight scale, which differ across systems (at 20% noise SGD+WD has weight norm 8.91 vs 23.71 for SGD), so we do not interpret it as intrinsic flatness.

VI. Ablations and Analysis

The two ablations (Table VII) and the sweeps of Sec. V were run with the same code, seeds and budget as the main group.

Random vs adversarial direction. Replacing the ascent direction by a random direction of the same radius removes the benefit entirely: on digits at 20% and 40% noise the accuracy is 0.800 and 0.623, against 0.799 and 0.625 for

TABLE II

Main results, final-epoch model, mean $\pm$ std over 5 seeds. gap_acc is noisy-train accuracy minus clean test accuracy; memorised is the fraction of flipped training labels fitted; sharp_adv is the loss increase after one normalised-gradient step of norm 0.5; sharp_rand is the mean loss increase over 20 random directions of the same norm, printed in scientific notation.

Method	Task	test_acc	gap_acc	memorised	sharp_adv	sharp_rand	weight_norm
SGD	digits_n0.0	0.959 ± 0.008	0.041 ± 0.008	0.000 ± 0.000	0.0019 ± 0.0005	$3.78 \times 10^{-5} \pm 9.2 \times 10^{-6}$	16.43 ± 0.09
SGD+WD	digits_n0.0	0.962 ± 0.008	0.038 ± 0.008	0.000 ± 0.000	0.0075 ± 0.0003	$1.39 \times 10^{-4} \pm 2.3 \times 10^{-5}$	12.99 ± 0.10
SAM	digits_n0.0	0.970 ± 0.006	0.030 ± 0.006	0.000 ± 0.000	0.0030 ± 0.0009	$9.71 \times 10^{-6} \pm 2.7 \times 10^{-6}$	17.41 ± 0.14
SGD	digits_n0.2	0.799 ± 0.020	0.200 ± 0.019	0.997 ± 0.007	0.0320 ± 0.0086	$1.13 \times 10^{-3} \pm 1.1 \times 10^{-4}$	23.71 ± 0.17
SGD+WD	digits_n0.2	0.931 ± 0.009	-0.147 ± 0.018	0.053 ± 0.032	0.0673 ± 0.0009	$1.80 \times 10^{-3} \pm 1.8 \times 10^{-4}$	8.91 ± 0.12
SAM	digits_n0.2	0.938 ± 0.013	-0.125 ± 0.025	0.156 ± 0.057	0.0585 ± 0.0032	$6.15 \times 10^{-4} \pm 8.0 \times 10^{-5}$	19.32 ± 0.37
SGD	digits_n0.4	0.625 ± 0.028	0.375 ± 0.028	1.000 ± 0.000	0.0331 ± 0.0115	$1.35 \times 10^{-3} \pm 1.4 \times 10^{-4}$	26.14 ± 0.20
SGD+WD	digits_n0.4	0.874 ± 0.011	-0.266 ± 0.025	0.098 ± 0.012	0.0736 ± 0.0071	$2.42 \times 10^{-3} \pm 2.3 \times 10^{-4}$	8.77 ± 0.20
SAM	digits_n0.4	0.901 ± 0.012	-0.276 ± 0.025	0.115 ± 0.031	0.0753 ± 0.0028	$5.39 \times 10^{-4} \pm 9.2 \times 10^{-5}$	16.75 ± 0.40
SGD	spirals_n0.2	0.919 ± 0.025	-0.146 ± 0.024	0.102 ± 0.022	0.0951 ± 0.0614	$1.11 \times 10^{-3} \pm 1.3 \times 10^{-4}$	11.98 ± 0.22
SGD+WD	spirals_n0.2	0.914 ± 0.028	-0.147 ± 0.031	0.088 ± 0.035	0.0465 ± 0.0144	$1.06 \times 10^{-3} \pm 1.5 \times 10^{-4}$	9.73 ± 0.21
SAM	spirals_n0.2	0.912 ± 0.024	-0.150 ± 0.030	0.085 ± 0.026	0.0420 ± 0.0136	$5.65 \times 10^{-4} \pm 1.0 \times 10^{-4}$	11.75 ± 0.19

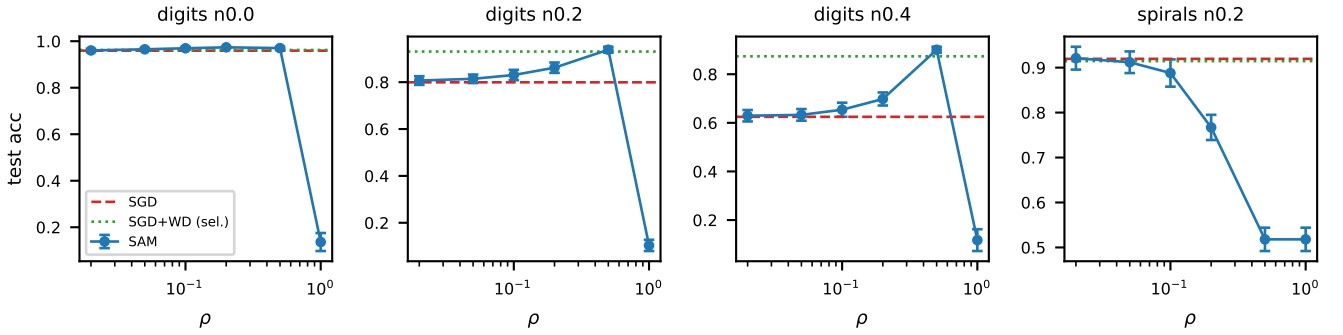


Fig. 1. Test accuracy (mean $\pm$ std, 5 seeds) versus ρ for SAM; dashed: SGD, dotted: SGD+WD at its selected λ . At $\rho = 1.0$ digits runs end at near-chance accuracy (task means 0.103–0.136).

TABLE III

Test accuracy of SAM minus baseline, with Welch and paired t -test p -values (5 seeds, uncorrected), and the Holm-adjusted paired p over the 8 rows.

task	vs	diff	Welch p	paired p	Holm p
digits_n0.0	SGD	0.010	0.056	7.31×10^{-3}	0.044
digits_n0.0	SGD+WD	0.0074	0.130	0.013	0.063
digits_n0.2	SGD	0.139	3.96×10^{-6}	8.75×10^{-5}	6.12×10^{-4}
digits_n0.2	SGD+WD	0.0078	0.294	0.049	0.148
digits_n0.4	SGD	0.276	1.98×10^{-6}	3.89×10^{-5}	3.11×10^{-4}
digits_n0.4	SGD+WD	0.027	7.30×10^{-3}	0.025	0.101
spirals_n0.2	SGD	-0.0075	0.644	0.320	0.640
spirals_n0.2	SGD+WD	-0.0025	0.884	0.606	0.640

TABLE IV

Test accuracy (mean over 5 seeds) of SAM by ρ (columns), SGD for reference.

task	SGD	0.02	0.05	0.1	0.2	0.5	1
digits_n0.0	0.959	0.960	0.965	0.970	0.974	0.970	0.136
digits_n0.2	0.799	0.807	0.814	0.831	0.862	0.938	0.103
digits_n0.4	0.625	0.629	0.633	0.654	0.698	0.901	0.118
spirals_n0.2	0.919	0.921	0.912	0.888	0.767	0.518	0.518

SGD, and nearly all flipped labels are memorised (0.998 and 1.000). With the same radius and the same extra

TABLE V

Test accuracy (mean over 5 seeds) of SGD+WD by weight decay λ .

task	0.0001	0.001	0.01	0.05
digits_n0.0	0.959	0.962	0.958	0.836
digits_n0.2	0.798	0.811	0.931	0.525
digits_n0.4	0.622	0.634	0.874	0.099
spirals_n0.2	0.918	0.914	0.677	0.518

compute, the gradient-aligned direction is therefore what produces the effect here, rather than weight noise of the same Euclidean norm or the extra step. This is a weak control: a random direction in the roughly 26k-dimensional parameter space changes the loss far less than the gradient direction (for SGD at 20% noise, a step of norm 0.5 raises the training loss by 1.13×10^{-3} in a random direction against 0.0320 along the gradient, Table II), so it does not rule out that larger random noise would help. On two-spirals and noise-free digits the random variant matches SGD too (0.919, 0.959). This isolates the perturbation direction, not the reason why the adversarial direction helps (e.g. clean-example up-weighting versus implicit regularisation [5]), which we did not test.

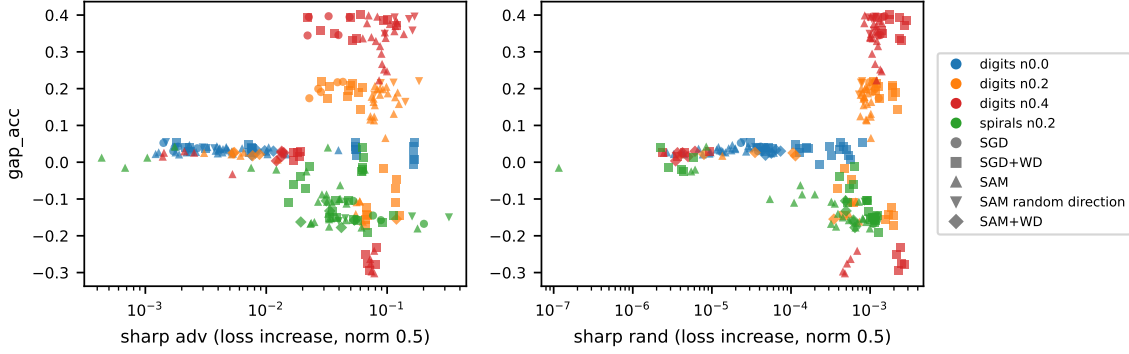


Fig. 2. Final-model sharpness (perturbation norm 0.5, noisy training set) versus gap_acc over all 260 evaluation runs (main, sweeps and ablation); colour: task, marker: system.

TABLE VI

Spearman correlation of sharpness (rand: random direction, adv: adversarial step) with the accuracy gap (acc) and loss gap (loss).

The pooled rows are filed in the run registry under the task digits_n0.4 because every registry row needs a task; they pool the tasks named in the row label.

subset	n	rand-acc	rand-loss	adv-acc	adv-loss
all tasks pooled	260	0.24	0.24	0.12	0.07
pooled, collapsed runs removed	219	0.27	0.24	0.15	0.09
digits_n0.0	65	0.13	-0.07	-0.20	-0.65
digits_n0.2	65	0.35	0.31	0.04	0.05
digits_n0.4	65	0.37	0.32	0.38	0.37
spirals_n0.2	65	-0.72	-0.85	-0.34	-0.40
digits_n0.0 no collapsed	62	0.08	-0.22	-0.16	-0.62
digits_n0.2 no collapsed	56	0.23	0.15	-0.15	-0.14
digits_n0.4 no collapsed	51	0.10	0.02	0.09	0.09
spirals_n0.2 no collapsed	50	-0.43	-0.69	-0.39	-0.54
SAM-sweep digits pooled	90	0.74	0.76	0.63	0.61
SAM only digits_n0.0	30	0.33	0.57	0.06	-0.46
SAM only digits_n0.2	30	0.56	0.56	0.73	0.71
SAM only digits_n0.4	30	0.56	0.56	0.51	0.51

SAM+WD (failure of naive combination). Combining the selected ρ and λ without joint tuning is harmful on noisy digits: test accuracy 0.223 ± 0.290 at 20% noise and 0.099 at 40% (chance; at 20% noise four seeds are near chance, the median is 0.094, and one reaches 0.741, so the mean is bimodal). In the collapsed runs the weights are driven towards zero (weight norm 0.32 ± 0.07 at 40% noise against 16.75 for SAM and 8.77 for SGD+WD). Without noise SAM+WD is comparable to SAM (0.972 ± 0.006 vs 0.970 ± 0.006), and on spirals it is 0.901 against 0.912 for SAM, within one seed standard deviation. On noisy digits each regulariser was selected alone at the grid value just below its own collapse (Tables IV,V), so together they may over-regularise; we did not run a joint grid, so we cannot say whether a jointly tuned SAM+WD would beat SAM or SGD+WD.

Sensitivity. Both regularisers show a cliff just above their best value on noisy digits (ρ : 0.938 to 0.103 from 0.5 to 1.0 at 20% noise; λ : 0.931 to 0.525 from 0.01 to 0.05). SAM’s best value gains only 0.0078 over SGD+WD’s at 20% noise. The practical reading is that at this scale a badly chosen ρ (or λ) can be much worse than plain SGD,

and that the comparison with a tuned weight decay is the relevant one.

VII. Limitations

Scale. One small architecture (3-layer MLP, width 128), tiny data (600 or 400 training points), two datasets, symmetric noise only, and 5 evaluation seeds: the paired and Welch tests have low power, the registered tests are uncorrected (with the Holm adjustment of Table III the SAM vs SGD+WD margin at 40% noise is no longer below 0.05), and several gaps (SAM vs SGD+WD at 20% noise) are within one standard deviation. The results do not transfer automatically to CNNs, large data or instance-dependent noise.

Tuning. The learning rate, batch size, epoch count and momentum were fixed for all systems and not tuned, so a differently tuned SGD or SGD+WD could shift the margins; ρ had 6 candidate values and λ 4, selection used a noisy validation split and 3 seeds, and for noisy digits the selected $\rho = 0.5$ sits next to a collapse at 1.0 so the true optimum is not resolved. On digits the tuning seeds re-split the same 1797 images as the evaluation seeds, so the images used for selection overlap the evaluation test images; this is the same for SAM and SGD+WD and only chooses one of 6 or 4 grid values, but the selection is not independent of the test data. SAM+WD was not tuned jointly. The comparison is epoch-matched, not compute-matched: SAM uses two gradient evaluations per step, and we did not run SGD or SGD+WD with twice the steps. We use the final-epoch model with no early stopping, whereas [5] report that SAM’s label-noise gains peak with early stopping; early-stopped SGD would be a natural, stronger baseline that we did not run.

Sharpness measurement. Our sharpness uses a single fixed Euclidean radius (0.5), one-step adversarial or 20 random directions, measured on the noisy training set. It is not scale invariant (cf. ASAM [8]), the weight norms differ widely across systems (at 40% noise 8.77 for SGD+WD against 26.14 for SGD), and the training

TABLE VII

Ablations at the selected ρ (and λ for SAM+WD), mean $\pm$ std over 5 seeds, same columns and units as Table II. Compare with SAM in Table II.

Method	Task	test_acc	gap_acc	memorised	sharp_adv	sharp_rand	weight_norm
SAM random direction	digits_n0.0	0.959 $\pm$ 0.009	0.041 $\pm$ 0.009	0.000 $\pm$ 0.000	0.0024 $\pm$ 0.0009	$3.78 \times 10^{-5} \pm 8.6 \times 10^{-6}$	16.44 $\pm$ 0.10
SAM+WD	digits_n0.0	0.972 $\pm$ 0.006	0.028 $\pm$ 0.006	0.000 $\pm$ 0.000	0.0090 $\pm$ 0.0016	$5.71 \times 10^{-5} \pm 1.2 \times 10^{-5}$	13.72 $\pm$ 0.14
SAM random direction	digits_n0.2	0.800 $\pm$ 0.025	0.200 $\pm$ 0.025	0.998 $\pm$ 0.003	0.1210 $\pm$ 0.0427	$7.79 \times 10^{-4} \pm 3.6 \times 10^{-5}$	23.74 $\pm$ 0.19
SAM+WD	digits_n0.2	0.223 $\pm$ 0.290	-0.013 $\pm$ 0.079	0.102 $\pm$ 0.020	0.0293 $\pm$ 0.0503	$1.22 \times 10^{-4} \pm 1.3 \times 10^{-4}$	1.61 $\pm$ 2.44
SAM random direction	digits_n0.4	0.623 $\pm$ 0.022	0.377 $\pm$ 0.022	1.000 $\pm$ 0.000	0.1387 $\pm$ 0.0317	$1.05 \times 10^{-3} \pm 8.8 \times 10^{-5}$	26.16 $\pm$ 0.19
SAM+WD	digits_n0.4	0.099 $\pm$ 0.005	0.020 $\pm$ 0.011	0.130 $\pm$ 0.011	0.0133 $\pm$ 0.0007	$4.50 \times 10^{-6} \pm 1.1 \times 10^{-6}$	0.32 $\pm$ 0.07
SAM random direction	spirals_n0.2	0.919 $\pm$ 0.024	-0.143 $\pm$ 0.021	0.097 $\pm$ 0.028	0.1232 $\pm$ 0.1235	$1.04 \times 10^{-3} \pm 1.4 \times 10^{-4}$	11.98 $\pm$ 0.24
SAM+WD	spirals_n0.2	0.901 $\pm$ 0.025	-0.147 $\pm$ 0.029	0.097 $\pm$ 0.033	0.0359 $\pm$ 0.0127	$5.72 \times 10^{-4} \pm 7.3 \times 10^{-5}$	9.49 $\pm$ 0.14

loss levels differ, so differences in S_{adv} across optimisers confound flatness with loss level and weight scale. The sharpness-gap correlations are observational, pool runs with different optimisers and some collapsed runs, and the exploratory “no collapsed” rows were defined after seeing the data. Our gap definition compares noisy-train and clean-test accuracy and is negative when a model does not memorise.

Implementation. All optimisers are reimplemented here from their published description, SAM uses a global-norm perturbation on every minibatch (not m-sharpness splits), and the spirals task is easy enough that no system memorises noise, which limits what it can show. A full study would add CNNs on CIFAR-N-style noise, more seeds, joint tuning, early stopping and scale-invariant sharpness.

VIII. Conclusion

On a small MLP with symmetric label noise, SAM improved clean test accuracy over plain SGD by a large margin (0.938 vs 0.799 and 0.901 vs 0.625 at 20% and 40% noise on digits), but weight decay alone recovered most of that gain, leaving a margin of 0.0078 at 20% noise (inconclusive) and 0.027 at 40% (both tests below 0.05 uncorrected with $n = 5$, one seed negative). SAM gave no benefit on noisy spirals, was sensitive to ρ , with a collapse to chance one grid step above the best value on noisy digits, and a random-direction perturbation of the same radius behaved like SGD. A fixed-radius loss-increase sharpness did not track the generalisation gap across optimisers (pooled Spearman 0.07–0.24, negative within spirals). Within the limits of a five-seed, epoch-matched CPU study with untuned learning rate, the practical take-away is to compare SAM against a tuned weight decay and to tune ρ on validation data; we make no claim about larger models.

References

[1] P. Foret, A. Kleiner, H. Mobahi, and B. Neyshabur, “Sharpness-Aware Minimization for Efficiently Improving Generalization,” arXiv preprint arXiv:2010.01412, 2020. [Online]. Available: <https://arxiv.org/abs/2010.01412>

[2] N. S. Keskar, D. Mudigere, J. Nocedal, M. Smelyanskiy, and P. T. P. Tang, “On Large-Batch Training for Deep Learning: Generalization Gap and Sharp Minima,” arXiv preprint arXiv:1609.04836, 2016. [Online]. Available: <https://arxiv.org/abs/1609.04836>

[3] Y. Jiang, B. Neyshabur, H. Mobahi, D. Krishnan, and S. Bengio, “Fantastic Generalization Measures and Where to Find Them,” arXiv preprint arXiv:1912.02178, 2019. [Online]. Available: <https://arxiv.org/abs/1912.02178>

[4] M. Andriushchenko, F. Croce, M. Müller, M. Hein, and N. Flammarion, “A Modern Look at the Relationship between Sharpness and Generalization,” arXiv preprint arXiv:2302.07011, 2023. [Online]. Available: <https://arxiv.org/abs/2302.07011>

[5] C. Baek, Z. Kolter, and A. Raghunathan, “Why is SAM Robust to Label Noise?” arXiv preprint arXiv:2405.03676, 2024. [Online]. Available: <https://arxiv.org/abs/2405.03676>

[6] M. Andriushchenko and N. Flammarion, “Towards Understanding Sharpness-Aware Minimization,” arXiv preprint arXiv:2206.06232, 2022. [Online]. Available: <https://arxiv.org/abs/2206.06232>

[7] C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals, “Understanding deep learning requires rethinking generalization,” arXiv preprint arXiv:1611.03530, 2016. [Online]. Available: <https://arxiv.org/abs/1611.03530>

[8] J. Kwon, J. Kim, H. Park, and I. K. Choi, “ASAM: Adaptive Sharpness-Aware Minimization for Scale-Invariant Learning of Deep Neural Networks,” arXiv preprint arXiv:2102.11600, 2021. [Online]. Available: <https://arxiv.org/abs/2102.11600>

[9] J. Du, H. Yan, J. Feng, J. T. Zhou, L. Zhen, R. S. M. Goh, and V. Y. F. Tan, “Efficient Sharpness-aware Minimization for Improved Training of Neural Networks,” arXiv preprint arXiv:2110.03141, 2021. [Online]. Available: <https://arxiv.org/abs/2110.03141>

[10] K. Wen, T. Ma, and Z. Li, “How Does Sharpness-Aware Minimization Minimize Sharpness?” arXiv preprint arXiv:2211.05729, 2022. [Online]. Available: <https://arxiv.org/abs/2211.05729>